

CORTICOMOTOR PLASTICITY FOLLOWING UNILATERAL STRENGTH TRAINING

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ABSTRACT: *Introduction:* We used transcranial magnetic stimulation (TMS) to investigate 3 weeks of unilateral leg strength training on ipsilateral motor cortex (iM1) excitability, and short-latency intracortical inhibition (SICI). *Methods:* Right leg dominant participants ($n = 14$) were randomly divided into either a strength training (ST) or control group. The ST group completed 9 training sessions (4 sets of 6 to 8 repetitions of single right leg squats). *Results:* We observed a 41% increase in right leg strength, and a 35% increase in strength of the untrained left leg ($P < 0.01$). There was a significant increase in motor evoked potential (MEP) amplitude recruitment curve for the untrained left leg ($P < 0.01$). SICI of the iM1 decreased by 21% for the untrained left leg ($P < 0.01$). *Conclusions:* The findings provide evidence for corticomotor adaptation for unilateral leg strength training within the iM1 that is modulated by changes in interhemispheric inhibition.

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It is well recognized that unilateral strength and skill training improves motor performance of not only the practiced limb, but also in the unpracticed contralateral homologous limb.^{1–5} Recent meta-analyses have shown that, on average, an 8% increase in strength of the untrained limb is observed following unilateral strength training.^{6,7} Neural adaptations have been implicated, because the cross transfer of strength occurs in the absence of muscle hypertrophy.⁷ Studies have shown that unilateral strength training leads to adaptive changes in the primary motor cortex ipsilateral to the training limb (iMI) in the form of increased corticospinal excitability, reduced short-latency intracortical inhibition (SICI), reduced interhemispheric inhibition (IHI) and increases in voluntary activation.^{4,8,9} In addition, several imaging (i.e., functional magnetic resonance imaging [fMRI], and positron emission tomography) and transcranial magnetic stimulation (TMS) studies have also demonstrated increased activation of the iMI and increased corticospinal excitability of the contralateral resting limb during unilateral voluntary iso-

metric contractions.^{10–14} Several studies have demonstrated increased neural drive from iMI to the untrained limb following maximal and submaximal unilateral strength training, suggesting that adaptations confined to the iMI may mediate the cross transfer of strength.^{8,9}

The role of intracortical inhibitory circuits in mediating the cross transfer of strength has not been examined comprehensively. Paired-pulse TMS techniques can be used to study intracortical circuitry of the M1. The application of a conditioning stimulus at a set interval activates interneuronal circuits within M1 that influence the excitability of corticospinal neurons to a subthreshold test stimulus delivered up to 20 ms later.^{15,16} The ability of the conditioning stimulus to inhibit (suppress) or facilitate (increase) the motor evoked potential (MEP) amplitude is dependent on the intensity of the conditioning stimulus (i.e., subthreshold or suprathreshold), and the length of the interstimulus interval (ISI).^{17,18} Suppression of the MEP with ISIs of 1–5 ms is due to activation of intracortical γ -aminobutyric acid-ergic (GABAergic) inhibitory interneurons and is termed SICI. TMS has been used to study SICI and IHI between motor cortices during the acquisition of motor skills.¹⁹ Recent cross-sectional data shows that SICI and IHI are reduced following unilateral skill and strength practice in both the trained and untrained limb.^{8,19} Furthermore, it has been shown recently that motor training with metronome-paced movement at a preferred tempo facilitates use-dependent plasticity.²⁰ Based upon these previous studies,^{8,19,20} it seems reasonable to hypothesize that unilateral strength training which incorporates controlled (i.e., metronome-paced) repetition timing may also induce changes in iMI excitability and inhibition (SICI) that may mediate the cross transfer of strength to the untrained homologous limb.

Therefore, the purpose of this study was to determine whether increases in corticomotor excitability and reductions in SICI of the iMI modulates the cross transfer of strength following heavy-load, controlled tempo strength training of the knee extensor muscles.

METHODS

Experimental Design. This study consisted of a between-participants repeated measures design, whereby individuals were randomly allocated to

Abbreviations: 1-RM, single repetition maximum; AMT, active motor threshold; IHI, interhemispheric inhibition; iM1, ipsilateral motor cortex; M1, primary motor cortex; MEP, motor evoked potential; MEP_{MAX}, maximum motor evoked potential; M_{MAX}, maximum compound wave (M-wave); MVIC, maximum voluntary isometric contraction; sEMG, surface electromyography; SICI, short-latency intracortical inhibition; ST, strength training group; TMS, transcranial magnetic stimulation

Key words: cross education, intracortical inhibition, primary motor cortex, strength training, transcranial magnetic stimulation

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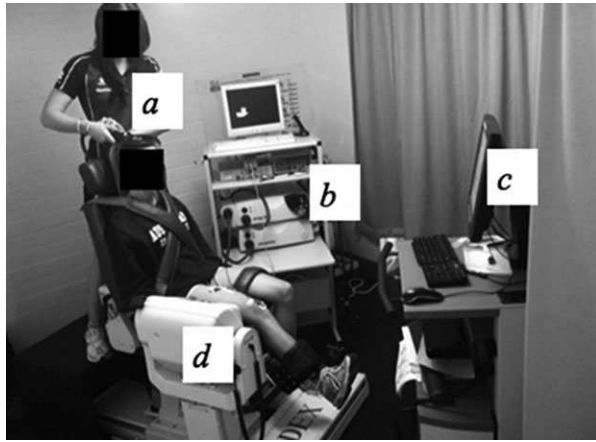


FIGURE 1. Visual representation of the experimental set up. (a) Placement of the circular coil. (b) The paired-pulse magstim. (c) The visual feedback on a computer monitor. (d) The isokinetic dynamometer.

either a strength training (ST) or a control group (no training). Participants assigned to ST completed 9 supervised strength training sessions over a 3-week period. Testing measures included single leg squat 1 repetition maximum (1-RM) strength and maximal voluntary isometric contraction (MVIC) torque (trained and untrained legs), muscle thickness assessment by means of ultrasound imaging, corticospinal excitability (recruitment curves), and SICI by means of single and paired pulse TMS (Fig. 1). Before the intervention, all participants undertook a familiarization session that involved TMS and maximal isotonic and isometric strength testing to ensure they were comfortable with both the testing and training procedures to minimize the potential effects of learning. All testing visits lasted approximately 60 min, and all training sessions were fully supervised and took approximately 20 min.

Participants. Fourteen healthy individuals recruited from the university student population (7 men and 7 women) aged between 18 and 35 years volunteered to participate in the study. All participants were right leg dominant²¹ and were systematically and randomly allocated according to gender and baseline strength to either a ST ($f = 3$; $m = 4$; 21 ± 1.1 years), or control group ($f = 4$; $m = 3$; 21 ± 1.2 years). All participants provided written informed consent before participation. All procedures were conducted according to the Helsinki Declaration of 1975 granted by the University Human Research Ethics Committee.

Maximum Strength Testing. Maximum voluntary dynamic strength of all participants was determined by a 1-RM single leg squat. All participants completed a warm-up that consisted of 5 min of moderate aerobic exercise on a cycle-ergometer,

and 2 warm-up sets of single leg squats with increasing weight. The 1-RM test involved performing single leg squats positioned under a power rack (Nautilus® Xpload™, Virginia, USA). Squat depth was determined by using an electromagnetic goniometer (3DM-GX2®, Williston, Vermont, USA) to control for knee joint angle (80°). The starting weight was determined by each participant's estimate of his or her leg strength.²² If the estimated weight was successful, the weight was then increased until the participant could no longer perform 1 repetition. The last successful trial was recorded as the 1-RM strength.²² Between each trial a 3-min rest period was allocated to minimize muscle fatigue.²² This procedure was performed for both legs. Additionally, isometric torque of the quadriceps was determined before and following the training intervention using an isokinetic dynamometer (Biodex system 4 Pro, Biodex Medical Systems, Shirley USA). Participants were placed in a seated position with a trunk-thigh angle of 110°. The axis of the dynamometer was then aligned with the anatomical axis of the knee joint, and the leg was held to the dynamometer lever arm using a padded strap positioned 1 cm superior to the malleoli of the ankle. To ensure that the trunk was stabilized during testing, a waist strap and 2 cross-over shoulder straps were used. During isometric testing, the knee was positioned at a 60° angle, and the participant was required to perform 3 maximal isometric leg extensions for 5 s with a 5-s rest period between each repetition. Verbal instructions and encouragement were provided to ensure that participants achieved their true MVIC torque. The highest peak torque of the 3 trials was taken and recorded as the participant's MVIC torque.

Measurement of Anterior Thigh Muscle Thickness. A Sonosite ultrasound (Springfield, NJ, USA) was used to measure muscle thickness of the anterior thigh before and after the training intervention. The site of measurement was determined by placing the transducer perpendicular to the long axis of the thigh on its superior aspect, three-fifths of the distance from the anterior superior iliac spine (ASIS) to the superior patella border.²³ A 6–15 Hz transducer probe was lubricated with transmission gel and placed lightly on the marked area. When a clear image was visible on the monitor, the pressure of the transducer was slowly reduced to ensure minimal compression of the muscle before the monitor was then frozen. To ensure accuracy of the data from pre to post testing, marking sites were recorded and matched on each testing session. Reliability trials were conducted for measurement of muscle thickness by ultrasonography.

Specifically, 8 participants were assessed on 2 consecutive days before the commencement of the intervention. An average of 6 recordings was obtained from each participant and subsequently used for reliability analysis. The testing procedure was found to be reliable, with no significant difference detected between the 2 trials and a coefficient of variation of 2.4% ($P = 0.66$; $r = 0.99$).

Transcranial Magnetic Stimulation and Electromyography. TMS was applied over the cortical representation of the quadriceps muscle group, using a circular coil (90-mm diameter) attached by means of a BiStim unit to two Magstim 200² stimulators (Magstim Co, UK). MEPs were produced by stimulation of the contralateral M1, innervating the trained and untrained legs.²⁴ The handle of the TMS coil was positioned over the vertex of the head and held tangential to the skull in an anterior–posterior orientation, so the current flowed in a counter clockwise direction for activating rectus femoris of the left and right legs. Sites near the estimated motor area of the quadriceps were explored and marked to determine the site in which the largest motor evoked potential (MEP) could be evoked during a low level contraction (10% MVIC). The active motor threshold (AMT) was determined as the minimum stimulus intensity required to elicit an MEP in the right and left rectus femoris of at least 200 μ V in 3 of 5 consecutive trials during low level voluntary knee extension (10% MVIC). AMT was expressed relative to 100% maximum stimulator output (MSO), and the stimulus intensity was altered in 1% increments of MSO throughout this process until the appropriate threshold level was achieved.

Surface electromyography (sEMG) was recorded from the rectus femoris muscle in both the trained and nontrained legs using bipolar Ag-AgCl electrodes. Two electrodes were positioned 2 cm apart on the mid belly of the rectus femoris muscle located between the ASIS and the patella, with the ground electrode placed on the bony prominence of the patella. All sEMG signals (MEPs) were sampled at 2048 Hz and collected on a PC running commercially-available software PowerLab 8 (ADInstruments, Australia) by means of a laboratory analogue-digital interface (PowerLab 8/30, ADInstrument, Australia) for later off-line analysis. sEMG signals were filtered and amplified ($\times 1000$) with bandpass filtering between 20 Hz and 500 Hz and digitized at 1.5 kHz for 500 ms, recorded and analyzed using PowerLab 8.

Recruitment Curves. The stimulus intensities used to establish the TMS recruitment curves were determined for each individual according to the AMT before the training intervention. At each stimulus intensity, 10 stimuli were applied over the

contralateral M1, with the percentage of stimulator output delivered in a pseudo-randomized manner. Two sets of 5 stimuli were given at stimulus intensity below each participant's AMT, increasing at 5% (of stimulator output) steps until no further increase in MEP amplitude was observed (MEP_{MAX}). Each stimulus was delivered in random intervals every 10 to 12 s to avoid stimulus anticipation, and 30 sec rest was provided between each set of stimuli to reduce the possibility of muscle fatigue.

Short-Latency Intracortical Inhibition. Short-latency intracortical inhibition (SICI) was assessed using a paired-pulse protocol that consisted of a subthreshold conditioning stimulus that preceded a supra-threshold test stimulus by 3 ms.²⁵ The test stimulus was set at 20% above AMT before and following the training intervention, whereas the conditioning stimulus was set at 70% of AMT. Fifteen test stimuli and 15 SICI stimuli were applied in a randomized order 12 s apart. The conditioning MEP amplitude was expressed as a percentage of the unconditioned test MEP amplitude to calculate the level of intracortical inhibition.

M-waves. Direct muscle responses were obtained from the right and left rectus femoris muscle by supramaximal electrical stimulation (pulse duration 1 ms) of the femoral nerve under resting conditions (DS7A, Digitimer, UK). The site of stimulation that produced the largest M-wave was located by positioning the bipolar electrodes in the femoral triangle lateral to the femoral artery, 3–5 cm below the inguinal ligament, under resting conditions. An increase in current strength was applied to the femoral nerve from below the participant's threshold until there was no further increase observed in the amplitude of the EMG response (M_{MAX}). To ensure maximal responses, the current was increased an additional 20%, and the average M_{MAX} was obtained from 5 stimuli delivered at 0.2 Hz.

Strength Training Procedures. Participants in the ST group undertook supervised unilateral strength training of the preferred (dominant) leg, 3 times per week for 3 weeks. The strength training program prescribed was progressively overloaded and periodized based on the maximum single leg squat strength of the dominant leg in the pretest session and then adjusted as necessary for the 3-week intervention. Before each session a 5-min warm-up was performed on a cycle ergometer at an intensity of 70% age-predicted maximum heart rate ($\pm 5\%$) to increase muscle temperature and blood flow. This was followed by 1 set at 12-RM and 1 set at 10-RM. Participants then completed their prescribed training. Participants completed a

prescribed training load of 4 sets of 80% of 1-RM (8 repetitions) in week 1, 82.5% of 1-RM (6–8 repetitions) in week 2 and 85.5% of 1-RM (6 repetitions) for week 3. Repetition timing was set at 3 s for the concentric phase and 4 s for the eccentric phase by means of the use of an electronic metronome. sEMG electrodes were placed on the rectus femoris muscle of the contralateral leg, which remained relaxed behind the participant and resting on a 20 cm box. Visual feedback of muscle activation was provided to the participant and investigator by means of an oscilloscope (HAMEG, Mainhausen, Germany) that was located 1 m in front of them at eye level to minimize muscle activation of the rested leg during training.

Contralateral Strength Transfer. The contralateral transfer of strength was calculated⁷ to determine the difference in change in the mean strength of the untrained nondominant leg in the control and trained groups following the training period. The calculation was performed as follows:

$$\left(\frac{E_{Post} - E_{Pre}}{E_{Pre}} - \frac{C_{Post} - C_{Pre}}{C_{Pre}} \right) 100$$

Where *E*_{Post} refers to mean posttraining 1-RM strength for the trained groups' untrained leg, *E*_{Pre} refers to mean pretraining 1-RM strength for the trained groups' untrained leg, *C*_{Post} refers to mean posttraining 1-RM strength for the controls' untrained (nondominant) leg, and *C*_{Pre} refers to mean pretraining 1-RM strength for the control groups' untrained (nondominant) leg.

Data Analysis. The root-mean-square (rms, μ V) of the pretrigger EMG in rectus femoris of the trained and untrained legs was determined over a 20 ms period before each TMS stimulus during pre- and posttesting. All MEP amplitudes were calculated from the peak-to-peak of the biphasic spike, averaged and normalized to the *M*_{MAX} and multiplied by 100. Recruitment curves were constructed by plotting stimulus intensity against MEP amplitude, and the data were fitted with a nonlinear Boltzmann sigmoid equation using Prism 5 (Graphpad software Inc., California, USA):

$$\text{MEP (s)} = \text{Bottom} + \frac{(\text{Top} - \text{Bottom})}{1 + \exp\left(\frac{V_{50} - X}{\text{Slope}}\right)}$$

where *s* represents stimulus intensity, Top represents the peak height, and *V*₅₀ represents the stimulus intensity at which the MEP amplitude is 50% of the MEP_{MAX} (half peak slope)

Sample Size Calculations and Statistical Analyses. The number of participants required was based upon power calculations for the expected

changes in mean rectified MEPs (sEMG recordings from the rectus femoris muscle). Using previous cross education data in healthy untrained adults,⁷ we estimated that 6 participants in each group would provide at least 80% power (95% confidence interval) to detect a 15% difference in mean rectified MEPs assuming a SD of 10–15% between groups at *P* < 0.05 (two-tailed).

Dependent variables (strength, pretrigger rmsEMG, MEP amplitude, recruitment curves and SICI) were analyzed using a 2 (time) by 2 (condition), two-way mixed-factor repeated measures ANOVA. Where appropriate, pair wise *post hoc* comparisons, with Bonferroni correction (*P* < 0.016), were used. Coefficient of variance (CoV) and paired *t*-tests were used to determine the reliability of the ultrasound testing protocol. Alpha was set at *P* ≤ 0.05, and all results are displayed as means ± SE.

RESULTS

Voluntary Muscle Strength. There was no difference in trained right leg 1-RM knee extensor strength between groups at baseline (*F*_{1,12} = 0.54; *P* = 1.00). Following the intervention, there was a main effect for time (*F*_{1,12} = 99.39; *P* < 0.001); however, there was no main effect for group (*F*_{1,12} = 1.14; *P* = 0.31). There was a group by time interaction following training (*F*_{1,12} = 80.32; *P* < 0.001). *Post hoc* analyses showed that heavy-load strength training resulted in a 40.7% increase in strength of the right knee extensors (*P* < 0.001). Similarly, at baseline, no differences in 1-RM strength of the untrained left knee extensors were present between groups (*F*_{1,12} = 0.14; *P* = 0.71). Following the intervention there was a main effect for time (*F*_{1,12} = 196.76; *P* < 0.001) but no main effect for group (*F*_{1,12} = 0.38; *P* = 0.55). A group by time interaction was detected following training (*F*_{1,12} = 183.86; *P* < 0.001). *Post hoc* analyses revealed there was a 35.4% increase in left knee extensor strength when compared with control (*P* < 0.001; Fig. 2).

There was a significant positive correlation between the percentage of strength gained in the trained right leg and the percentage of the contralateral transfer of strength to the untrained left leg (*r*² = 0.74; *P* < 0.001; Fig. 3). Unilateral strength training of the right leg resulted in a 34.8% strength-transfer to the contralateral untrained left leg.

Muscle Thickness. There were no differences in muscle thickness of the right trained rectus femoris between the groups at baseline (*F*_{1,12} = 3.28; *P* = 0.10). There was no main effect for time (*F*_{1,12} = 3.24; *P* = 0.10), group (*F*_{1,12} = 0.85; *P* = 0.37) or group by time interactions (*F*_{1,12} = 1.00; *P* = 0.34)

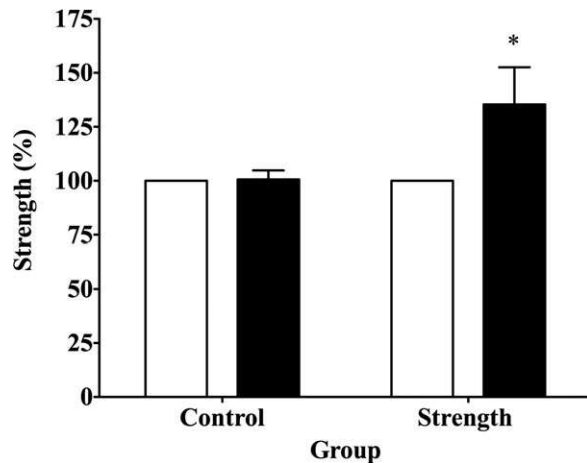


FIGURE 2. Mean ($\pm$ SE) 1-RM knee extensor strength of the left untrained leg (expressed as a mean percentage change) for the strength training and control groups following heavy-load unilateral strength training of the right leg. Light bars represent pretraining strength and dark bars represent posttraining strength. There was a 35.4% increase in left knee extensor strength for the strength training (ST) group following the intervention. Asterisk (*) denotes a significant increase in knee extensor strength from pre- to posttraining between groups ($P < 0.001$).

detected following training. There were also no differences in muscle thickness of the left untrained rectus femoris between groups at baseline ($F_{1,12} = 3.43$; $P = 0.09$). There was a main effect for time ($F_{1,12} = 5.19$; $P = 0.04$); however, no main effect for group ($F_{1,12} = 1.02$; $P = 0.33$) or group by time interactions was detected for the left leg following the intervention ($F_{1,12} = 1.87$; $P = 0.20$).

rmsEMG. Mean ($\pm$ SE) values (μ V) for rmsEMG 20 ms before TMS stimulation at 10% MVIC torque, during pre- and posttesting were obtained. For the trained right leg, no main effects for time ($F_{1,12} = 0.16$; $P = 0.70$), group ($F_{1,12} = 5.87$; $P = 0.33$) or group by time interactions ($F_{1,12} = 0.01$; $P = 0.91$) were detected. Similarly, for the untrained left leg no main effects for time ($F_{1,12} = 0.01$; $P = 0.93$) group ($F_{1,12} = 0.65$; $P = 0.44$) or group by time interactions ($F_{1,12} = 0.01$; $P = 0.92$) were present following strength training. Furthermore, there were no differences in mean ($\pm$ SE) muscle activation (expressed as μ V) of the untrained rectus femoris during training over the 3 week intervention (week 1, 5.89 ± 1.94 ; week 2, 5.03 ± 3.94 ; week 3, 6.57 ± 1.82 ; $P = 0.92$).

Active Motor Threshold and Motor Evoked Potentials. Mean ($\pm$ SE) TMS stimulus intensity for the trained right leg at AMT before training was $46.6 \pm 0.04\%$ and $51.9 \pm 0.04\%$ for control and strength training groups respectively. For the untrained left leg before training, TMS stimulus intensity was

$47.0 \pm 0.02\%$ and $49.5 \pm 0.05\%$ for control and strength training groups, respectively. Following the training intervention, TMS stimulus intensity at AMT for the untrained left leg was $46.0 \pm 0.02\%$ and $44.3 \pm 0.04\%$, while the trained right leg was $45.7 \pm 0.02\%$ and $49.0 \pm 0.04\%$ for control and strength training groups respectively (Table 1).

Mean ($\pm$ SE) absolute values for MEP amplitude at AMT (expressed as a percentage of M_{MAX}) for the trained right and untrained left leg, pre- and postintervention are displayed in Table 1. For the trained right leg, no differences were present in MEP amplitude at AMT at baseline ($F_{1,11} = 0.54$; $P = 0.47$). There were also no main effects for time ($F_{1,11} = 0.39$; $P = 0.55$), group ($F_{1,11} = 0.19$; $P = 0.67$) or group by time interactions following training ($F_{1,11} = 1.87$; $P = 0.20$). Similarly, for the untrained left leg, there were no differences in MEP amplitude at AMT at baseline between groups ($F_{1,12} = 5.49$; $P = 0.43$). There was no main effect for time ($F_{1,12} = 0.21$; $P = 0.66$) or group ($F_{1,12} = 1.25$; $P = 0.20$) following training. Furthermore, no group by time interactions were detected ($F_{1,12} = 0.142$; $P = 0.72$).

For the trained right leg, no differences in MEP amplitude at 20% above AMT (expressed as a percentage of M_{MAX}) were detected between groups at baseline ($F_{1,11} = 1.06$; $P = 0.33$). Following the intervention, there was a main effect for time ($F_{1,11} = 22.73$; $P = 0.001$), however, there was no main effect for group ($F_{1,11} = 1.5$; $P = 0.24$). In addition, following training, there was a group by time interaction ($F_{1,11} = 26.06$; $P < 0.001$), and *post hoc* analyses showed MEP amplitude at 20% above AMT increased by 55.4% following training ($P < 0.001$). For the untrained left leg, there were

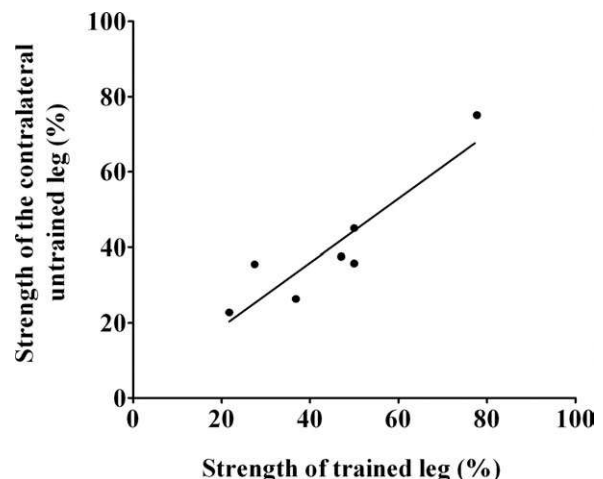


FIGURE 3. Strength changes (represented by Δ) for knee extensor 1-RM strength for the trained and untrained contralateral leg in trained participants following 3 weeks of heavy-load unilateral strength training. Data are expressed as a percentage of pretraining strength ($r^2 = 0.74$; $P < 0.001$).

Table 1. . Mean ($\pm$ SE) for neurological variables of the right and left leg for all groups..

		Stimulator output at AMT (%)		MEP amplitude at AMT (% M_{MAX})		MEP amplitude at 20% above AMT (% M_{MAX})		MEP _{MAX} (% M_{MAX})		SICI (% test response)	
	Group	Pre	Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Untrained left leg	Control	47.0	46.0	13.22	13.33	34.94	34.72	60.77	62.57	36.16	35.33
	ST	± 0.02	± 0.02	± 2.05	± 2.14	± 3.96	± 4.00	± 5.98	± 7.59	± 3.97	± 3.80
Trained right leg		49.5	44.3	19.3	20.15	35.35	57.38*	52.66	71.07*	43.24	34.05*
		± 0.05	± 0.04	± 4.48	± 3.03	± 5.95	± 6.24	± 5.76	± 4.88	± 5.14	± 3.91
	Control	46.6	45.7	17.84	16.79	45.97	45.39	62.16	61.66	35.73	35.35
		± 0.02	± 0.02	± 1.94	± 1.77	± 4.59	± 4.67	± 3.49	± 2.56	± 6.72	± 6.62
	ST	51.9	49.0	15.63	18.67	30.26	47.02*	34.34	53.88*	40.02	30.22*
		± 0.04	± 0.04	± 2.79	± 3.13	± 2.44	± 4.30	± 4.84	± 5.21	± 5.13	± 1.97

Stimulator output at AMT (%), MEP amplitude at AMT (% of M_{MAX}), MEP amplitude at 20% above AMT (% of M_{MAX}), MEP_{MAX} (% of M_{MAX}), and SICI.

*Indicates a significant group by time effect ($p < 0.05$). ST, strength training group; AMT, active motor threshold; MEP, motor evoked potential; MMAX, maximum M-wave; MEP_{MAX}, maximum motor evoked potential; SICI, short-latency intracortical inhibition.

no differences between groups in MEP amplitude at 20% above AMT at baseline ($F_{1,12} = 3.14$; $P = 0.10$). Following the intervention, there was a main effect for time ($F_{1,12} = 33.02$; $P = 0.001$); however, no main effect for group ($F_{1,12} = 2.82$; $P = 0.12$).

Training resulted in a group by time interaction ($F_{1,12} = 21.42$; $P = 0.001$), and *post hoc* analyses showed a 62.3% increase in MEP amplitude at 20% above AMT for the strength training group compared with the control ($P = 0.003$; Fig. 4).

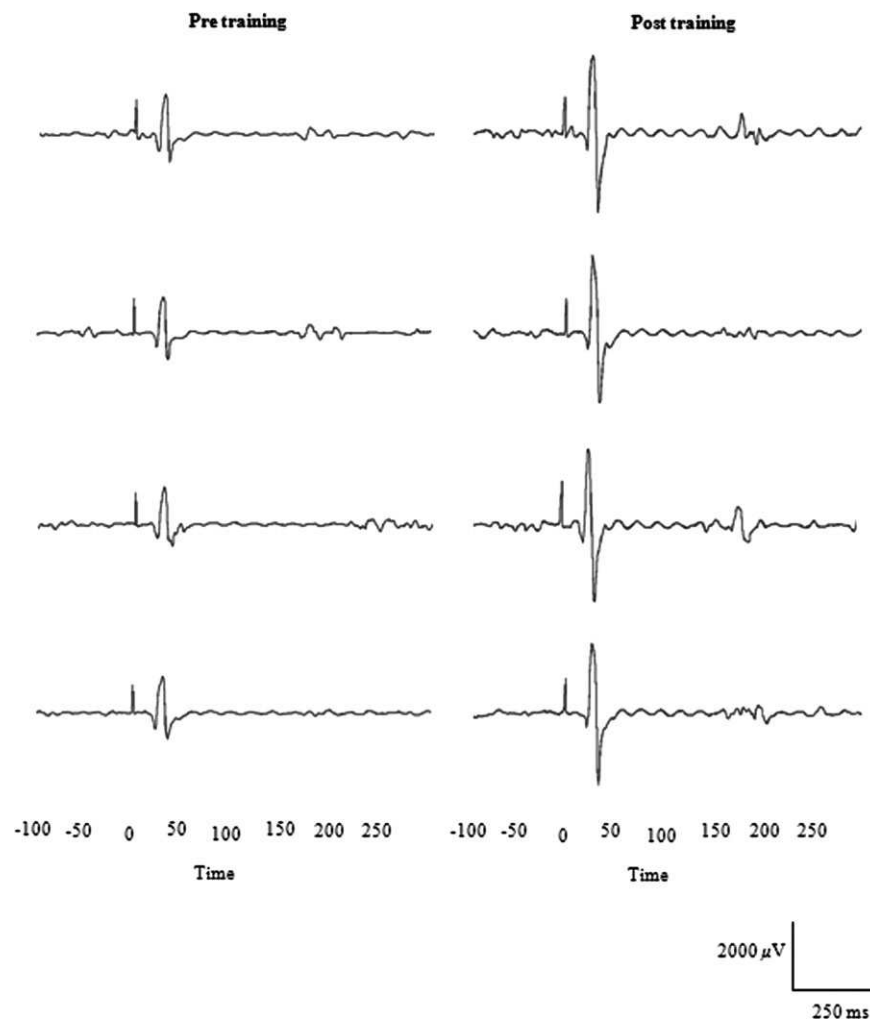


FIGURE 4. Motor evoked potentials (MEPs) for the untrained left rectus femoris in 1 participant from the strength training (ST) group, pre (left) and post (right) intervention. MEP sweeps were recorded at 20% above AMT.

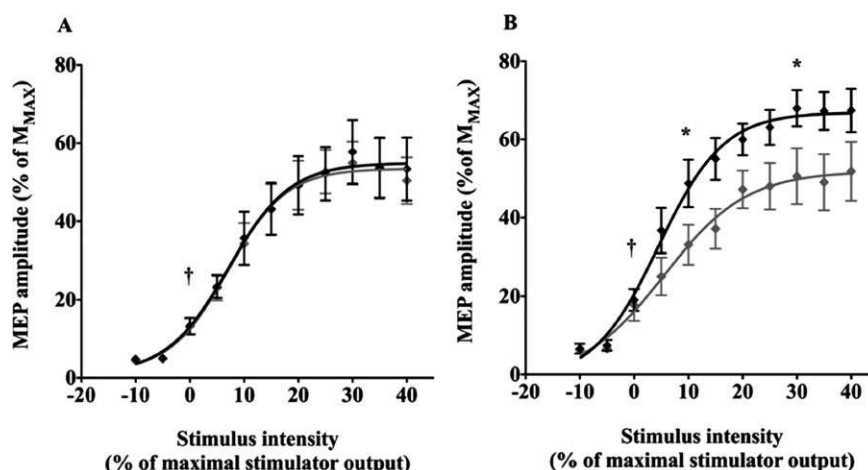


FIGURE 5. Mean ($\pm$ SE) motor evoked potential (MEP) amplitudes (expressed as a percentage of M_{MAX}) obtained from the left untrained knee extensors of the (a) Control and (b) ST groups. The smallest to largest MEPs are responses to stimulation at -10 , 0 , 10 , 20 , 30 , and 40% of maximal stimulator output above active motor threshold (AMT). Each stimulus-response curve is characterized by the minimal stimulus intensity required to elicit MEPs (AMT), estimated peak slope, maximum level of MEP amplitude (MEP_{MAX}) and the stimulus intensity at which the MEP amplitude is 50% of the MEP_{MAX} ($V50$). Light curves represent pre-training data, and dark curves represent posttraining data. Following training, there were increases in both MEP amplitude at 20% above AMT and MEP_{MAX} for the ST group ($P < 0.016$). † identifies AMT. Asterisk (*) denotes significant increases at 20% above AMT and peak height.

For the trained right leg, there were no differences in MEP_{MAX} (expressed as a percentage of M_{MAX}) between groups at baseline ($F_{1,10} = 0.02$; $P = 0.89$). Following the intervention there was a main effect for time ($F_{1,10} = 10.87$; $P = 0.008$), group ($F_{1,10} = 11.37$; $P = 0.007$) and a group by time interaction ($F_{1,10} = 12.06$; $P = 0.006$). *Post hoc* analyses revealed that strength training increased MEP_{MAX} by 56.9% compared with the control group ($P = 0.001$). For the untrained left leg, there were also no differences between groups in MEP_{MAX} at baseline ($F_{1,10} = 0.37$; $P = 0.56$). Following the intervention, there was a main effect for time ($F_{1,11} = 26.57$; $P < 0.001$), however, there was no main effect for group ($F_{1,11} = 0.01$; $P = 0.98$). A group by time interaction was detected following the intervention ($F_{1,11} = 17.92$; $P = 0.001$). *Post hoc* analyses revealed a 34.9% increase in MEP_{MAX} following strength training when compared with the control group ($P < 0.001$; Fig 5B).

Recruitment Curves. Recruitment curves were constructed to determine the values for peak height, half peak slope ($V50$) and the estimated peak slope for the trained right leg and untrained left leg of each participant pre- and posttraining. There was a group by time interaction effect for peak height in the trained right leg following the intervention ($F_{1,8} = 12.84$; $P = 0.007$). *Post hoc* analyses revealed a 52.9% increase in peak height following training when compared with the control ($P = 0.007$). For the untrained left leg, there was a group by time interaction effect for peak height following the intervention ($F_{1,12} = 19.85$; $P = 0.001$; Fig. 5A and B), and *post hoc* analyses showed

a 29.1% increase in peak height following training ($P = 0.001$). There were no main effects for time, group, or group by time interactions for either leg for $V50$ or estimated peak slope following training ($P > 0.05$).

Short-latency Intracortical Inhibition. For the trained right leg, no differences in SICI between groups were detected at baseline ($F_{1,11} = 1.29$; $P = 0.28$). There was a significant main effect for time ($F_{1,11} = 6.20$; $P = 0.03$), however, there was no main effect for group ($F_{1,11} = 0.003$; $P = 0.96$). There was a group by time interaction following training ($F_{1,11} = 5.30$; $P = 0.04$), and *post hoc* analyses revealed a 24.5% reduction in SICI ($P = 0.004$) when compared with the control. For the untrained left leg, there were no differences in SICI between groups at baseline ($F_{1,12} = 2.20$; $P = 0.16$). A main effect for time was present ($F_{1,12} = 16.29$; $P = 0.002$), however, there was no main effect for group ($F_{1,12} = 0.48$; $P = 0.50$). Following training, there was a group by time interaction effect ($F_{1,12} = 12.46$; $P = 0.004$; Fig. 6), and *post hoc* analyses showed a 21.3% decrease in SICI ($P = 0.004$) following strength training when compared with the control.

DISCUSSION

The purpose of this study was to explore the contralateral effects of unilateral strength training and to ascertain the potential mechanism mediating the cross transfer of strength. We observed that following 3 weeks of heavy-load controlled unilateral strength training, there was increased dynamic strength of the homologous muscle of the untrained contralateral knee extensors accompanied by a

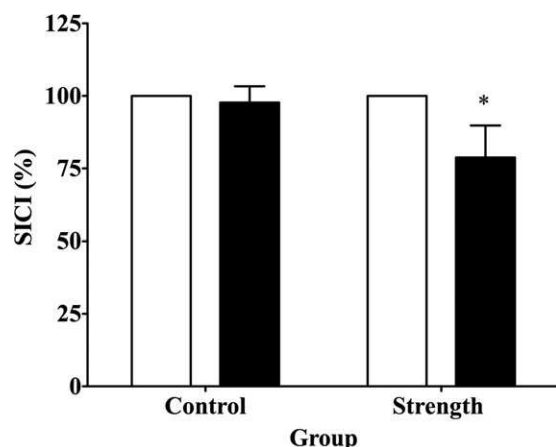


FIGURE 6. Mean ($\pm$ SE) short-latency intracortical inhibition (SICI; expressed as a mean percentage change) evoked by paired pulse transcranial magnetic stimulation (TMS) pre- (light bars) and posttraining (dark bars) in the left untrained knee extensors for the strength training (ST) and control groups. Following training, there was a 21.3% reduction in SICI in the ST group when compared with the control ($P = 0.004$). Asterisk (*) denotes a significant reduction in SICI following training.

decrease in SICI and increased corticomotor excitability confined to the iMI. These results provide further evidence that changes in cortical inhibition and excitability are involved in the cross transfer of strength.

Increased Voluntary Force Following Unilateral Strength Training. The increases in strength observed in the trained and untrained legs are comparable to other general cross education studies.^{4,5,8,26,27} Hortobágyi et al.²⁶ demonstrated a 30%, 22%, and 77% increase in concentric, isometric and eccentric voluntary force production respectively following lower limb unilateral strength training, which also supports our findings of a 35% increase in left untrained knee extensor strength for the ST group. The contralateral transfer of strength observed in this study (35%) is greater than recent investigations that have used a similar between-subjects design and have reported a contralateral strength transfer of 7%⁹ and 9%.⁵ The large difference in the cross transfer of strength between this study and that of previous studies may be due in part to the methodology of the strength training used. In 1 of the few studies to use isotonic training, Shima et al.²⁷ demonstrated an 18.9% increase in strength in the plantar flexor muscles, with a 7.8% increase in the contralateral limb, after 6 weeks training of 4 days per week (3×10 – 12 repetitions at 70–75% of 1-RM). A contributing factor that may account for the greater strength gains in our study, particularly in the untrained limb, may lie in the exercise intensity prescribed (being of a higher intensity in the majority of studies). An aspect to this study was

that the training load prescribed to novice participants commenced at 80% of their 1-RM, while other studies prescribed progressive overload training at moderate intensities (see review by Carroll et al.⁷).

Changes in Corticomotor Plasticity Following Unilateral Strength Training. To date, limited research has used paired-pulse TMS to explore the corticomotor changes following unilateral strength training. The findings from this study are in general agreement with TMS studies that have shown increased activation of iMI and increased excitability of the corticospinal tract targeting the contralateral resting limb during strong unilateral contractions.^{12,14,28} Furthermore, with the recent finding of increased cortical activation following unilateral strength training,⁹ our data suggest that the cross transfer of strength is mediated by short-term changes in iMI excitability and inhibition. Given that high force unilateral voluntary contractions have been shown to affect the efficacy of neural circuits controlling the untrained limb,^{12,29,30} the results suggest that the strength training program used in this study increased the neural excitability of the contralateral homologous muscle due to chronic changes in synaptic connectivity within specific neural circuits between hemispheres that contribute to the ability to generate force.^{8,19,28} We also tested the effect of unilateral strength training on the functional properties (construction of recruitment curves) of the corticospinal tract following training in the iMI. Recruitment curves provide important information about the spatial distribution of the excitable elements of the M1 and corticospinal tract. In accordance with our previous cross education study,⁴ we have demonstrated functional changes in the corticospinal tract ipsilateral to the training limb. The change in the recruitment curve (at 20% above AMT and peak height) in the iMI to the trained leg is consistent to what has been reported in the same cortical area following motor skill training.³¹ Taken together, our findings are consistent with the understanding that modulation of corticospinal excitability within and between the MIs has likely contributed to the cross transfer of strength following heavy-load controlled strength training. Although not directly measured, it is likely that the interactions between the 2 cerebral hemispheres contributed to the cross transfer of strength to the untrained leg, potentially through mechanism associated with reduced IHI.⁸

A novel finding of this study was the significant reduction in SICI in the iMI. Several studies support the opinion that GABA-mediated SICI contributes to the acquisition of new skills.^{19,28}

SICI has been shown to be reduced in contralateral hand muscles following both strength and skill practice.^{19,28} Consistent with our findings, several recent studies have demonstrated reduced SICI in the iMI following contralateral motor skill practice in both upper and lower limb muscles.^{31,32} Furthermore, given that previous studies have demonstrated task-dependant changes in intracortical inhibition following skilled tasks,^{1,33,34} the changes in SICI observed in this study may also be a result of task-dependant changes that are similar to those observed with motor skill acquisition.^{19,31} It appears that the cross transfer of strength has altered the balance between excitation and inhibition confined to the iMI, therefore improving voluntary motor drive due to the reduced intracortical inhibition.

Several studies show that complex interhemispheric connections (i.e., transcallosal pathways) from the M1 may provide a medium for neural drive to stimulate the opposite contralateral homologous muscle during unilateral contraction.^{7,29,35} The term “motor irradiation” describes the bilateral corticospinal activity that occurs during unilateral movement, which is likely to be mediated by such pathways.^{29,35} Current literature indicates that motor irradiation is a candidate mechanism that may mediate the cross transfer of strength observed following unilateral training.^{36–39} Furthermore, cortical activity in the passive limb appears to be dependent on the magnitude of excitation directed to the moving limb,^{35,36,39} and it occurs at a greater magnitude when voluntary contraction takes place in the dominant limb.³⁵ Our results show that repetitive strong unilateral contractions provide additional excitation to the iMI by means of functional changes in transcallosal pathways (i.e., reduced IHI) and intracortical inhibitory circuits (reduced SICI) that result in a training-related response to control corticospinal output from iM1.³⁶ This conclusion is in line with recent investigations that have shown that SICI generates diffuse cortical inhibition that affects corticospinal output and transcallosal pathways to a similar extent.^{28,37} Taken together, these findings suggest that there has been a reduction in the synaptic activity between inhibitory interneurons in the target hemisphere (i.e., iMI) that has attenuated the descending corticospinal output, thus modulating the cross transfer of strength.^{8,31}

In recent years, attempts have been made to identify the primary site of adaptation that contributes to the modulation of corticospinal excitability following unilateral motor training. Potential mechanisms could involve changes in synaptic strength, strengthening of existing corticospinal cells, and shifting the strength in functional con-

nectivity between hemispheres.³⁸ Adjustments in synaptic efficacy are likely to reflect enhanced postsynaptic potentials within the iMI and corticospinal tract as demonstrated by the facilitated MEPs in the recruitment curve. Several studies have reported increased excitability in supraspinal structures following high-intensity motor training.^{39–41} Certainly, our findings support a cortical mechanism (i.e., reduced SICI) as opposed to a spinal mechanism in modulating the cross transfer of strength.

Although IHI is thought to modulate the excitatory and inhibitory balance between the 2 cerebral hemispheres, a limitation of this study was the inability to directly quantify and assess the functional changes mediated by transcallosal pathways. Nevertheless, the reduction in SICI confined to the iMI is an important finding, which demonstrates its role in the modulation of cross education.

These findings have important clinical implications for patient populations that have reduced capacity to use 1 limb. For example, it has been shown that the use of unilateral strength training may attenuate the loss of strength and neural activation of an immobilized limb.⁴² Furthermore, our results show that unilateral strength training modulates the balance of iMI excitability and inhibition, which has clinical implications for certain neurological conditions, such as stroke. Future research should focus on the clinical application of unilateral strength training in an injured/or diseased population.

In conclusion, these findings suggest that the cross transfer of strength following unilateral strength training may be modulated by functional changes in excitability between the motor cortices. In particular, the data demonstrate that reduced SICI has attenuated the output from iMI to the untrained limb, suggesting that intracortical inhibition confined to the iMI may mediate the cross transfer of strength.

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